

Esercitazione semantica

1.
$$\left. \begin{array}{l} d_1 \text{ var } x: \text{bool} = \text{true}; \\ d_2 \text{ const } y: \text{bool} = \text{false}; \\ c \text{ } x := x \text{ and } y; \end{array} \right\} d = d_1; d_2 \left. \vphantom{\begin{array}{l} d_1 \text{ var } x: \text{bool} = \text{true}; \\ d_2 \text{ const } y: \text{bool} = \text{false}; \\ c \text{ } x := x \text{ and } y; \end{array}} \right\} d; c$$

$\rho = \emptyset \quad \sigma = \emptyset$

$$\frac{\emptyset \vdash \langle d, \sigma \rangle \rightarrow^* \langle \rho, \sigma_1 \rangle \quad \rho \vdash \langle c, \sigma_1 \rangle \rightarrow \sigma_2}{\emptyset \vdash \langle d; c, \sigma \rangle \rightarrow^* \sigma_2}$$

Elaborazione della dichiarazione

1.

$$\begin{array}{l} \frac{\emptyset \vdash \langle d_1, \sigma \rangle \rightarrow \langle \rho_1, \sigma_1 \rangle}{\emptyset \vdash \langle d_1; d_2, \sigma \rangle \rightarrow \langle \rho_1; d_2, \sigma_1 \rangle} \\ \rightarrow \emptyset \vdash \langle \text{var } x: \text{bool} = \text{true}, \sigma \rangle \rightarrow \langle [x \mapsto \text{true}], [x \mapsto \text{true}] \rangle \\ \downarrow \\ \frac{\emptyset \vdash \langle \text{var } x: \text{bool} = \text{true}, \sigma \rangle \rightarrow \langle [x \mapsto \text{true}], [x \mapsto \text{true}] \rangle}{\emptyset \vdash \langle d_1; d_2, \sigma \rangle \rightarrow \langle \underbrace{[x \mapsto \text{true}]}_{\rho_1}; d_2, \underbrace{[x \mapsto \text{true}]}_{\sigma_1} \rangle} \end{array}$$

2.

$$\begin{array}{l} \frac{\rho_1 \vdash \langle \text{const } y: \text{bool} = \text{false}, \sigma_1 \rangle \rightarrow \langle \underbrace{[y \mapsto \text{false}]}_{\rho_2}, \sigma_1 \rangle}{\emptyset \vdash \langle \rho_1; \text{const } y: \text{bool} = \text{false}, \sigma_1 \rangle \rightarrow \langle \rho_1; \rho_2, \sigma_1 \rangle} \end{array}$$

3.

$$\begin{array}{l} \emptyset \vdash \langle \rho_1; \rho_2, \sigma_1 \rangle \rightarrow \langle \rho_1[\rho_2], \sigma_1 \rangle \rightarrow \langle \underbrace{[x \mapsto \text{true}, y \mapsto \text{false}]}_{\rho}, \sigma_1 \rangle \\ \xrightarrow{*} \\ \langle d_1; d_2, \sigma \rangle \rightarrow \langle \rho_1; d_2, \sigma_1 \rangle \rightarrow \langle \rho_1; \rho_2, \sigma_1 \rangle \rightarrow \langle \rho, \sigma_1 \rangle \end{array}$$

Elaborazione dell'assegnamento

$$\sigma_1 = [x \mapsto \text{true}]$$

$$\rho = [x \mapsto \text{true}, y \mapsto \text{false}]$$

1.

$$\frac{\rho \vdash \langle x \text{ and } y, \sigma_1 \rangle \rightarrow \langle t, \sigma_1 \rangle}{\rho \vdash \langle x := x \text{ and } y, \sigma_1 \rangle \rightarrow \langle x := t, \sigma_1 \rangle}$$

$$\frac{P \vdash \langle x, \sigma_1 \rangle \rightarrow \langle \text{true}, \sigma_1 \rangle}{P \vdash \langle x \text{ and } y, \sigma_1 \rangle \rightarrow \langle \text{true and } y, \sigma_1 \rangle} \quad \sigma_1(P(x)) = \text{true}$$

2.

$$\frac{P \vdash \langle y, \sigma_1 \rangle \rightarrow \langle \text{false}, \sigma_1 \rangle}{P \vdash \langle \text{true and } y, \sigma_1 \rangle \rightarrow \langle \text{true and false}, \sigma_1 \rangle} \quad \sigma_1(y) = \text{false}$$

3.

$$\langle \text{true and false}, \sigma_1 \rangle \rightarrow \langle \overbrace{\text{false}}^t, \sigma_1 \rangle$$

Riscriviamo la regola di assegnamento

$$\frac{P \vdash \langle x \text{ and } y, \sigma_1 \rangle \rightarrow \langle \text{false}, \sigma_1 \rangle}{P \vdash \langle x := x \text{ and } y, \sigma_1 \rangle \rightarrow \langle x := \text{false}, \sigma_1 \rangle}$$

$$P \vdash \langle x := \text{false}, \sigma_1 \rangle \rightarrow \underbrace{\sigma_1 [l_x \rightarrow \text{false}]}_{\sigma_2} \wedge P(x) = l_x$$

Concludendo

$$\emptyset \vdash \langle d; c, \sigma \rangle \rightarrow \sigma_2$$

$$2 \quad p = [y \mapsto 2y, z \mapsto 4] \quad \sigma = [2y \mapsto -2]$$

if $y \geq 0$ then skip else $y := z - 1$;

Regola per esecuzione dell'if

$$p \vdash \langle y \geq 0, \sigma \rangle \rightarrow^* \langle t, \sigma \rangle$$

$$\frac{}{p \vdash \langle \text{if } y \geq 0 \text{ then } c_1 \text{ else } c_2, \sigma \rangle \rightarrow \langle \text{if } t \text{ then } \dots, \sigma \rangle}$$

Valutiamo la guardia $y \geq 0$

1.

$$\frac{p \vdash \langle y, \sigma \rangle \rightarrow \langle -2, \sigma \rangle}{p \vdash \langle y \geq 0, \sigma \rangle \rightarrow \langle -2 \geq 0, \sigma \rangle} \quad \sigma(p(y)) = -2$$

$$2. \quad \langle -2 \geq 0, \sigma \rangle \rightarrow \langle \underbrace{\text{false}}_t, \sigma \rangle$$

Riscriviamo la regola dell'if

$$p \vdash \langle y \geq 0, \sigma \rangle \rightarrow^* \langle \text{false}, \sigma \rangle$$

$$\frac{}{p \vdash \langle \text{if } y \geq 0 \text{ then } c_1 \text{ else } c_2, \sigma \rangle \rightarrow \langle \text{if } \text{false} \text{ then } \dots, \sigma \rangle}$$

$$p \vdash \langle \text{if } \text{false} \text{ then skip else } y := z - 1, \sigma \rangle \rightarrow \langle y := z - 1, \sigma \rangle$$

Eseguiamo l'assegnamento

$$\frac{\frac{p \vdash \langle z - 1, \sigma \rangle \rightarrow^* \langle n, \sigma \rangle}{p \vdash \langle y := z - 1, \sigma \rangle \rightarrow \langle y := n, \sigma \rangle}}{p \vdash \langle z, \sigma \rangle \rightarrow \langle 4, \sigma \rangle} \quad p(z) = 4$$

$$p \vdash \langle z - 1, \sigma \rangle \rightarrow \langle 4 - 1, \sigma \rangle$$

$$p \vdash \langle 4 - 1, \sigma \rangle \rightarrow \langle \underbrace{3}_n, \sigma \rangle$$

Riscriviamo l'assegnamento

$$\frac{p \vdash \langle z - 1, \sigma \rangle \rightarrow^* \langle 3, \sigma \rangle}{p \vdash \langle y := z - 1, \sigma \rangle \rightarrow \langle y := 3, \sigma \rangle}$$

